

PIERRE 24 S, SD, USA

WMO: 726630

Lat: 44.0194N

Lon: 100.3530W

Elev: 2124

StdP: 13.60

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 94085

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-4.5	-16.1	2.5	-8.8	-11.1	3.3	-3.9	29.7	17.4	26.9	19.9	8.1	N/A	0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.3	95.5	69.9	91.3	69.4	87.7	68.4	74.8	87.4	72.8	84.9	71.0	82.9	8.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.2	124.4	80.7	69.0	115.3	79.3	67.4	108.8	77.8	39.7	87.9	37.9	85.4	36.3	82.6	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	20.5	17.7	-15.9	102.1	3.9	4.5	-18.7	105.3	-21.0	107.9	-23.2	110.5	-26.0	113.8
			-16.5	77.7	4.7	2.0	-19.9	79.1	-22.6	80.3	-25.3	81.5	-28.7	83.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	20.8	21.0	35.5	44.9	56.3	67.2	74.5	71.8	63.9	48.6	35.7	22.4	(6)
	DBStd	21.77	12.94	13.83	13.93	10.83	8.72	6.54	6.12	6.02	8.28	10.04	11.67	12.24	(7)
	HDD50	3908	906	815	478	215	39	1	0	0	6	148	443	857	(8)
	HDD65	7315	1371	1233	916	604	292	51	5	12	118	513	878	1322	(9)
	CDD50	2818	0	2	27	63	235	515	759	675	421	105	16	0	(10)
	CDD65	752	0	0	1	2	23	116	299	222	83	6	0	0	(11)
	CDH74	8214	0	0	24	76	319	1171	3279	2247	981	115	2	0	(12)
	CDH80	3486	0	0	3	17	90	398	1580	984	376	38	0	0	(13)
(14) Wind	WSAvg	7.8	9.4	8.4	8.9	9.3	8.3	6.4	5.2	6.3	7.2	8.3	8.0	8.3	(14)
(15) Precipitation	PrecAvg	19.0	0.4	0.5	0.9	2.0	3.1	3.6	2.4	2.0	1.6	1.5	0.6	0.4	(15)
	PrecMax	26.9	2.0	1.3	3.0	4.1	6.0	7.5	5.9	5.4	5.7	4.5	1.6	1.2	(16)
	PrecMin	8.6	0.0	0.0	0.0	0.3	0.6	1.0	0.3	0.1	0.0	0.2	0.0	0.0	(17)
	PrecStd	4.2	0.4	0.4	0.7	1.1	1.5	1.6	1.5	1.2	1.3	1.4	0.5	0.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	64.2	77.7	81.8	88.1	94.3	103.8	98.9	96.2	85.5	71.9	56.8	(19)
		MCWB	42.4	48.0	53.5	57.5	65.9	69.7	69.8	69.1	65.3	59.6	51.3	43.7	(20)
	2%	DB	46.3	51.3	69.0	74.7	82.0	88.2	97.2	93.8	88.9	75.9	63.6	48.7	(21)
		MCWB	37.9	41.3	49.2	54.8	63.5	70.1	71.1	69.4	65.5	56.5	48.1	38.5	(22)
	5%	DB	41.6	44.9	62.5	68.7	77.1	84.5	92.6	89.4	84.0	69.9	57.7	42.6	(23)
		MCWB	35.2	37.8	47.2	51.4	61.3	67.8	70.8	69.1	64.4	53.2	44.8	35.1	(24)
	10%	DB	38.2	39.9	55.6	63.2	72.3	81.0	88.6	85.7	79.7	64.4	52.3	38.4	(25)
		MCWB	33.5	34.5	44.5	48.2	58.7	66.8	69.5	67.9	62.2	50.8	41.9	32.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	48.3	55.4	59.1	69.0	75.1	78.3	76.0	71.8	62.1	52.8	44.5	(27)
		MCDB	54.9	64.9	66.9	76.7	82.9	88.6	90.6	90.2	84.1	81.1	70.4	56.1	(28)
	2%	WB	38.6	43.3	50.8	55.4	66.2	72.8	75.7	73.5	68.9	58.2	48.5	39.3	(29)
		MCDB	46.2	54.6	68.0	71.9	78.3	84.1	87.5	85.8	81.4	73.4	62.9	48.7	(30)
	5%	WB	35.8	38.6	47.9	52.4	63.3	70.7	73.6	71.4	66.8	55.2	45.1	36.0	(31)
		MCDB	41.2	45.7	62.2	66.4	73.7	81.6	85.9	82.4	79.0	68.0	56.4	42.8	(32)
	10%	WB	33.8	35.3	44.5	49.6	60.2	68.5	71.7	69.5	64.6	52.2	41.9	33.3	(33)
		MCDB	38.5	40.9	56.3	61.2	69.6	79.1	84.6	80.5	76.3	63.9	51.1	38.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	18.4	21.0	22.2	21.5	21.6	24.3	23.6	24.3	21.2	20.2	16.6	(35)
		MCDBR	23.0	25.0	32.4	33.6	28.5	27.3	30.0	29.6	30.7	31.5	28.4	22.9	(36)
		MCWBR	15.7	16.2	17.3	18.0	13.8	13.3	10.6	11.3	12.1	15.8	16.1	14.6	(37)
	5% WB	MCDBR	21.8	24.3	31.5	31.0	25.0	24.6	25.2	24.7	26.3	28.7	27.4	21.8	(38)
		MCWBR	15.8	15.9	16.9	17.4	13.1	13.4	11.6	11.6	12.3	15.0	15.7	14.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.258	0.267	0.289	0.332	0.352	0.365	0.383	0.378	0.342	0.303	0.269	0.256	(40)
	taud		2.533	2.525	2.527	2.396	2.421	2.411	2.392	2.384	2.462	2.544	2.552	2.524	(41)
	Ebn at Noon		284	299	304	296	291	286	280	277	281	281	277	273	(42)
	Edh at Noon		21	25	29	36	36	36	37	36	30	24	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		607	907	1245	1569	1885	2133	2194	1839	1473	974	657	496	(44)
	RadStd		49	74	98	114	179	141	85	135	98	116	63	33	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page